Exercise 5

Ex. No.: 5

Perform Web Server Reconnaissance using Various Tools

Aim:

To understand and perform web server reconnaissance using different tools to gather information

about a target server, its services, and configurations.

Tools:

Kali Linux / Windows with tools installed

Tools:

- `whois`
- `nslookup` / `dig`
- `nmap`
- `nikto`
- `whatweb` / `wappalyzer`
- Burp Suite / OWASP ZAP

WHOIS Lookup

Purpose: Find domain registration details, owner, registrar, and name servers.

DNS Enumeration (nslookup / dig)

Purpose: Discover DNS records like A, MX, TXT, and subdomains.

Dig command

Port Scanning with NMAP: Identify open ports, services, and versions

Web Server Vulnerability Scan with Nikto

Purpose: Find common vulnerabilities and misconfigurations.

Technology Fingerprinting with WhatWeb

Purpose:Identify technologies running on the web server.

Wappalyzer Crome Extension: install Wappalyzer browser extension and visit the site

Result:

Collected domain and DNS information.

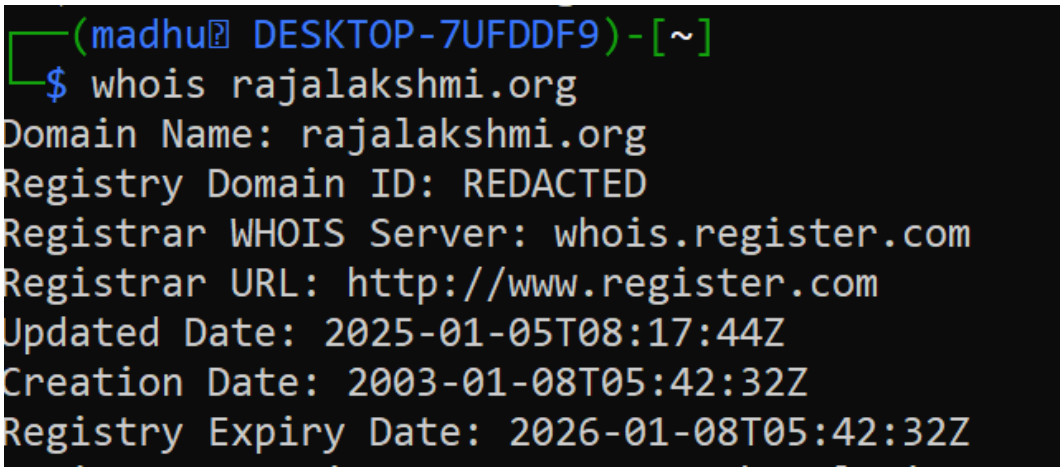
Identified open ports and services.

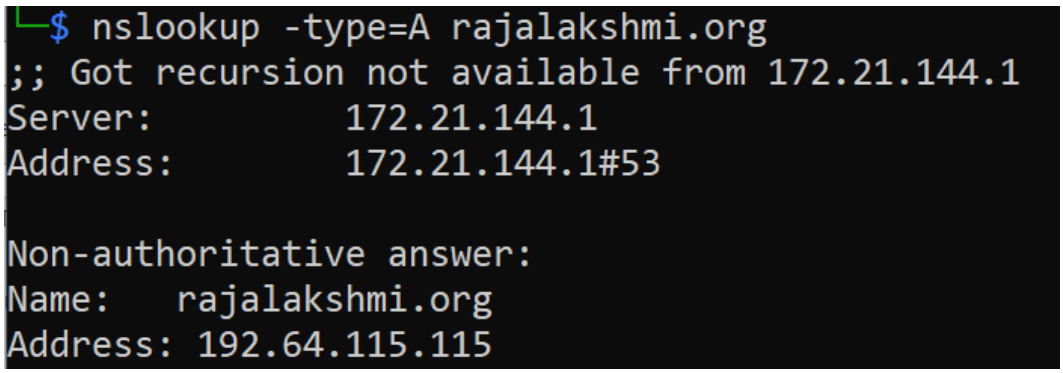
Fingerprinted web server technologies.

Discovered server misconfigurations and vulnerabilities.

Understood practical use of reconnaissance tools.

Screenshots





```
└─$ dig rajalakshmi.org

; <<>> DiG 9.20.9-1-Debian <<>> rajalakshmi.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22476
;; flags: qr rd ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;rajalakshmi.org.               IN      A

;; ANSWER SECTION:
rajalakshmi.org.        0       IN      A       192.64.115.115

;; Query time: 8 msec
;; SERVER: 172.21.144.1#53(172.21.144.1) (UDP)
;; WHEN: Sat Aug 23 13:01:49 IST 2025
;; MSG SIZE  rcvd: 64
```

```
└─$ nmap -sV rajalakshmi.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-23 13:03 IST
Nmap scan report for rajalakshmi.org (192.64.115.115)
Host is up (0.23s latency).
Not shown: 988 closed tcp ports (reset)
PORT     STATE    SERVICE      VERSION
21/tcp   open     tcpwrapped
22/tcp   open     ssh          OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0
25/tcp   filtered smtp
80/tcp   open     http         Golang net/http server
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     ssl/http     Golang net/http server
445/tcp  filtered microsoft-ds
3011/tcp open     trusted-web?
5900/tcp filtered vnc
8010/tcp open     http         Uvicorn
8011/tcp open     http         Uvicorn
```

```
└─$ nikto -h https://rajalakshmi.org
- Nikto v2.5.0
---------------------------------------------------------------------------
+ Target IP:          192.64.115.115
+ Target Hostname:    rajalakshmi.org
+ Target Port:        443
---------------------------------------------------------------------------
+ SSL Info:        Subject:  /CN=rajalakshmi.org
                   Altnames: rajalakshmi.org
                   Ciphers:  TLS_CHACHA20_POLY1305_SHA256
                   Issuer:   /C=US/O=Let's Encrypt/CN=R13
+ Start Time:         2025-08-23 13:11:05 (GMT5.5)
---------------------------------------------------------------------------
+ Server: No banner retrieved
+ /: Retrieved x-powered-by header: Next.js.
+ /: Uncommon header 'x-nextjs-prerender' found, with contents: 1.
+ /: Uncommon header 'x-nextjs-cache' found, with contents: HIT.
+ /: Uncommon header 'x-nextjs-stale-time' found, with contents: 300.
+ /: An alt-svc header was found which is advertising HTTP/3. The endpoint is: ':443'. Nikto cannot test HTTP/3 over
UIC. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/alt-svc
+ /WGP5H91S.cobalt: Drupal Link header found with value: </_next/static/media/4f05ba3a6752a328-s.p.woff2>; rel=preloa
; as="font"; crossorigin=""; type="font/woff2". See: https://www.drupal.org/
+ /WGP5H91S.php#: The X-Content-Type-Options header is not set. This could allow the user agent to render the content
of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulner
abilities/missing-content-type-header/
```

```
└─$ whatweb rajalakshmi.org
http://rajalakshmi.org [302 Found] Country[UNITED STATES][US], IP[192.64.115.115], RedirectLocation[https://rajalakshm
i.org/], Via-Proxy[HTTP/1.1 forward.http.proxy:3128]
https://rajalakshmi.org/ [200 OK] Country[UNITED STATES][US], Email[admin@rajalakshmi.org], HTML5, IP[192.64.115.115],
 Meta-Author[Rajalakshmi Engineering College], Open-Graph-Protocol, Script, Strict-Transport-Security[max-age=31536000
; includeSubDomains], Title[Rajalakshmi Engineering College - Premier Engineering Institution], UncommonHeaders[alt-sv
c,referrer-policy,x-content-type-options,x-dns-prefetch-control,x-nextjs-cache,x-nextjs-prerender,x-nextjs-stale-time]
, X-Frame-Options[DENY], X-Powered-By[Next.js]
```

